**Algorithm 1** Meta-evolution

**Inputs:** baseline; $G$ generations; $O$ offspring per generation; $P$ population size; tasks $T$;
$N_{\text{init}}$ initial seeds; $B$ TTTS seeds per generation; $N_{\text{reeval}}$ uniform reevaluation seeds; $N_{\text{val}}$ validation seeds;
reevaluation `strategy` $\in \{$`TTTS`, `uniform`$\}$.

`population` $\leftarrow$ [`baseline`]
`archive` $\leftarrow \{\}$
**for** $g = 1, \ldots, G$ **do**
  **for** $o = 1, \ldots, O$ **do** ▷ *Sample offspring*
    `parents` $\sim$ SampleParents(`population`)
    `prompt` $\sim$ Prompt($T$, `parents`)
    $x_o \sim$ LLM(`prompt`)
    $\hat{\mu}_{x_o} \leftarrow$ Evaluate($x_o, T, N_{\text{init}}$)
    $n_{x_o} \leftarrow N_{\text{init}}$
    `archive` $\leftarrow$ `archive` $\cup \{x_o\}$
  **if** `strategy` = `TTTS` **then** ▷ *Reevaluate*
    $\psi_x \leftarrow$ top-two Thompson probability of $x$ under $\mathcal{N}(\hat{\mu}_x, \hat{\sigma}^2/n_x)$
    Allocate $B$ seeds by sampling members of `archive` with replacement $\propto \psi_x$
  **else if** `strategy` = `uniform` **then**
    Allocate $N_{\text{reeval}}$ seeds to each new member of `population`
  Evaluate each member on its allocated seeds and update $\hat{\mu}_x$, $n_x$
  `population` $\leftarrow$ MapElites(`archive`) ▷ *One elite per task*
  $x_g^* \leftarrow \arg\max_{x \in \texttt{population}} \hat{\mu}_x$ ▷ *Validate generation's best*
  `val`$[x_g^*] \leftarrow$ Evaluate($x_g^*, T_{\text{val}}, N_{\text{val}}$)
**return** $\arg\max_x$ `val`$[x]$